	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	575 - gemm_wmma_kernel	(8, 8, 1)x(128, 4, 1)	65.73 us	106,349	0 - NVIDIA GeForce RTX 4060 Laptop GPU	1.62 Ghz	[19425] gemm_tensor	

Summary
Details
Source
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GPU Speed Of Light Throughput

Roofline Tensor Core

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	39.00	Duration [us]	65.73
Memory Throughput [%]	29.15	Elapsed Cycles [cycle]	106,349
L1/TEX Cache Throughput [%]	30.25	SM Active Cycles [cycle]	102,393.92
L2 Cache Throughput [%]	10.62	SM Frequency [Ghz]	1.62
DRAM Throughput [%]	9.39	DRAM Frequency [Ghz]	7.99

Small Grid This kernel grid is too small to fill the available resources on this device, resulting in only 0.9 full waves across all SMs. Look at Launch Statistics for more details.

Key Performance Indicators

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved close to 0% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

Tensor Core Operations Roofline

10
1
0.1
Performance [OP/s] (1 = 1E+13)
1
10
100
1,000
10,000
100,000
Arithmetic Intensity [OP/byte]

	# Operations	# Operations / Cycle	# Operations / s	Peak %	Peak Operations / Cycle	Peak Operations / s
Src:bf16 Dst:fp32 Sparsity:off	0	0	0	0	12,288	19,865,861,733,203.51
Src:bf16 Dst:fp32 Sparsity:on	0	0	0	0	24,576	39,731,723,466,407.02
Src:fp16 Dst:fp16 Sparsity:off	0	0	0	0	24,576	39,731,723,466,407.02
Src:fp16 Dst:fp16 Sparsity:on	0	0	0	0	49,152	79,463,446,932,814.03
Src:fp16 Dst:fp32 Sparsity:off	268,435,456	2,526.17	4,084,035,053,554.04	20.56	12,288	19,865,861,733,203.51
Src:fp16 Dst:fp32 Sparsity:on	0	0	0	0	24,576	39,731,723,466,407.02
Src:fp64	0	0	0	0	86.54	139,900,434,740.87
Src:fp8 Sparsity:off	0	0	0	0	49,152	79,463,446,932,814.03
Src:fp8 Sparsity:on	0	0	0	0	98,304	158,926,893,865,628...
Src:int1	0	0	0	0	393,216	635,707,575,462,512...
Src:int4 Sparsity:off	0	0	0	0	98,304	158,926,893,865,628...
Src:int4 Sparsity:on	0	0	0	0	196,608	317,853,787,731,256...
Src:int8 Sparsity:off	0	0	0	0	49,152	79,463,446,932,814.03
Src:int8 Sparsity:on	0	0	0	0	98,304	158,926,893,865,628...
Src:tf32 Dst:fp32 Sparsity:off	0	0	0	0	6,144	9,932,930,866,601.75
Src:tf32 Dst:fp32 Sparsity:on	0	0	0	0	12,288	19,865,861,733,203.51

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	1	# Pass Groups	2
Maximum Buffer Size [Mbytes]	4	-	-

0s
ms
+0.02ms
+0.04ms
+0.06ms
+0.08ms
+0.1ms
+0.12ms
+0.14ms
+0.16ms
Overview
Average Active Warps Per Cycle
42.7 warp
0
Total Active Warps Per Cycle
1.02k warp
0
Blocks Launched
64 block
0
SM Active Cycles
1.63k cycle
0
Executed Ipc Active
...6 inst/cycle
0
SM
SM ALU Pipe Throughput
100 %
0
SM FMA Light Pipe Throughput
100 %
0
SM FMA Heavy Pipe Throughput
100 %
0
SM Tensor Pipe Throughput
100 %
0
SM XU Pipe Throughput
100 %
0
DRAM
DRAM Throughput
100 %
0
DRAM Read Bandwidth
100 %
0
DRAM Write Bandwidth
100 %
0
L2 Cache
L2 Throughput
100 %
0
L2 Hit Rate
100 %
0
L1 Cache
L1 Throughput
100 %
0
L1 Writeback Throughput
478 cycle
0
L1 Hit Rate
100 %
0
L1 Wavefronts (Data)
1.21k
0
Workload Execution
gemm_wmma_kernel

Compute Workload Analysis

Pipe Utilization (Elapsed Cycles)

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	1.56	SM Busy [%]	39.00
Executed Ipc Active [inst/cycle]	1.62	Issue Slots Busy [%]	39.00
Issued Ipc Active [inst/cycle]	1.62		

Balanced ALU is the highest-utilized pipeline (32.1%) based on elapsed cycles in the workload, taking into account the rates of its different instructions. It executes integer and logic operations. It is well-utilized, but should not be a bottleneck.

Pipe Utilization (% of elapsed cycles)

0.0
25.0
50.0
75.0
100.0
ALU
FMA
Tensor (All)
Tensor (FP)
FP64
Tensor (INT)
Utilization [%]

Pipe Utilization (% of peak instructions executed over elapsed cycles)

0.0
25.0
50.0
75.0
100.0
ALU
LSU
FMA
ADU
Tensor (FP)
CBU
Uniform
XU
FMA (FP16)
FP64 (DMMA)
FP64 (FP64)
TEX
Tensor (INT)
Utilization [%]

Memory Workload Analysis

Memory Chart

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/s]	24.01	Mem Busy [%]	28.11
L1/TEX Hit Rate [%]	34.18	Max Bandwidth [%]	29.15
L2 Hit Rate [%]	79.72	Mem Pipes Busy [%]	29.15
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	0
L2 Compression Success Rate [%]	0	L2 Persisting Size [Mbyte]	6.29

L1TEX Global Load Access Pattern
Est. Speedup: 1.62%
The memory access pattern for global loads from L1TEX might not be optimal. On average, only 30.2 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the Source Counters section for uncoalesced global loads.

Key Performance Indicators

L1TEX Global Store Access Pattern
Est. Speedup: 14.58%
The memory access pattern for global stores to L1TEX might not be optimal. On average, only 16.0 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the Source Counters section for uncoalesced global stores.

Key Performance Indicators

Memory Chart

Values: Transfer Size
Inactivity: Greyed Out

Kernel
139.26 K Inst
Global
135.17 K Req
4.10 K Req
0.00 Inst
Local
0.00 Req
0.00 Req
0.00 Inst
Texture
0.00 Req
0.00 Inst
Surface
0.00 Req
0.00 Req
0.00 Inst
Load Global Store Shared
196.61 K Inst
Shared
65.54 K Req
131.07 K Req
L1/TEX Cache Hit Rate: 34.18 %
0.00 B
0.00 B
Shared Memory
6.69 MB
747.71 KB
L2 Cache Hit Rate: 79.72 %
0.00 B
0.00 B
L2 Compression Ratio: 0.00
16.90 KB
0.00 B
0.00 B
System Memory
1.58 MB
0.00 B
Device Memory
0.00 B
0.00 B
Peer Memory
2.72 K Req
ICC Hit Rate: 91.90 %
336.00 Req
832.51 K Req
IMC Hit Rate: 99.66 %
408.00 Req
1.02 K Req
IDC Hit Rate: 86.33 %
GCC Hit Rate: 74.19 %
% Peak
100%
80%
60%
40%
20%
0%

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	10.41	No Eligible [%]	59.23
Eligible Warps Per Scheduler [warp]	1.04	One or More Eligible [%]	40.77
Issued Warp Per Scheduler	0.41		

Issue Slot Utilization
Est. Local Speedup: 59.23%
Every scheduler is capable of issuing one instruction per cycle, but for this workload each scheduler only issues an instruction every 2.5 cycles. This might leave hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 12 warps per scheduler, this workload allocates an average of 10.41 active warps per scheduler, but only an average of 1.04 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, avoid possible load imbalances due to highly different execution durations per warp. Reducing stalls indicated on the Warp State Statistics and Source Counters sections can help, too.

Key Performance Indicators

Warps Per Scheduler

GPU Maximum Warps Per Scheduler
Theoretical Warps Per Scheduler
Active Warps Per Scheduler
Eligible Warps Per Scheduler
Issued Warp Per Scheduler
0.0
4.0
8.0
12.0
16.0

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	25.53	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	25.59	Avg. Not Predicated Off Threads Per Warp	29.57

Long Scoreboard Stalls
Est. Speedup: 44.30%
On average, each warp of this workload spends 11.3 cycles being stalled waiting for a scoreboard dependency on a L1TEX (local, global, surface, texture) operation. Find the instruction producing the data being waited upon to identify the culprit. To reduce the number of cycles waiting on L1TEX data accesses verify the memory access patterns are optimal for the target architecture, attempt to increase cache hit rates by increasing data locality (coalescing), or by changing the cache configuration. Consider moving frequently used data to shared memory. This stall type represents about 44.3% of the total average of 25.5 cycles between issuing two instructions.

Key Performance Indicators

Warp Stall Check the Warp Stall Sampling (All Samples) table for the top stall locations in your source based on sampling data. The Profiling Guide provides more details on each stall reason.

Warp State (All Cycles)

0.0
5.0
10.0
15.0
Warp States
Stall Long Scoreboard
Stall Barrier
Stall Wait
Stall Math Pipe Throttle
Stall Not Selected
Stall Short Scoreboard
Selected
Stall Dispatch Stall
Stall MIO Throttle
Stall Branch Resolving
Stall IMC Miss
Stall No Instruction
Stall Drain
Stall LG Throttle
Stall Misc
Stall Tex Throttle
Stall Membar
Stall Sleeping
Cycles per Instruction

Instruction Statistics

Opcode Category Chart

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	3,970,048	Avg. Executed Instructions Per Scheduler [inst]	41,354.67
Issued Instructions [inst]	3,978,901	Avg. Issued Instructions Per Scheduler [inst]	41,446.89

FP32 Non-Fused Instructions
Est. Speedup: 3.65%
This kernel executes 8192 fused and 8192 non-fused FP32 instructions. By converting pairs of non-fused instructions to their fused, higher-throughput equivalent, the achieved FP32 performance could be increased by up to 25% (relative to its current performance).

Key Performance Indicators

Executed Instruction Categories

0.0
2,000,000.0
4,000,000.0
Opcode Categories
Integer
Control
Load/Store
Movement
Miscellaneous
Floating Point
Uniform Datapath
Conversion
Executed Warp-Level Instructions/Opcode

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

NUMA ID Table

GPU ID	GPU Name	NUMA ID by CPU Affinity	CPU Affinity	NUMA ID by Memory Affinity
0	NVIDIA GeForce RTX 4060 Laptop GPU	0	0-15	0

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	64	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	28	Static Shared Memory Per Block [Kbyte/block]	5.38
Block Size	512	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	32,768	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	0.89	Shared Memory Configuration Size [Kbyte]	65.54
Uses Green Context	0	Stack Size	1,024
# SMs [SM]	24	# TPCs	12
Enabled TPC IDs	all	-	-

Occupancy

% Occupancy Graphs

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	4
Theoretical Active Warps per SM [warp]	48	Block Limit Shared Mem [block]	10
Achieved Occupancy [%]	87.05	Block Limit Warps [block]	3
Achieved Active Warps Per SM [warp]	41.78	Block Limit SM [block]	24

Achieved Occupancy
Est. Speedup: 12.95%
The difference between calculated theoretical (100.0%) and measured achieved occupancy (87.1%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

Key Performance Indicators

Impact of Varying Register Count Per Thread

125
100
75
50
25
0
% Occupancy
0
8
16
24
32
40
48
56
64
72
80
88
96
104
112
120
128
136
144
152
160
168
176
184
192
200
208
216
224
232
240
248
256
Registers Per Thread

Impact of Varying Block Size

125
100
75
50
25
0
% Occupancy
0
32
64
96
128
160
192
224
256
288
320
352
384
416
448
480
512
544
576
608
640
672
704
736
768
800
832
864
896
928
960
992
1,024
Block Size

Impact of Varying Shared Memory Usage Per Block

125
100
75
50
25
0
% Occupancy
0
3,184
6,368
9,552
12,736
15,920
19,104
22,288
25,472
28,656
31,840
35,024
38,208
41,392
44,576
47,760
50,944
54,128
57,312
60,496
63,680
66,864
70,048
73,232
76,416
79,600
82,784
85,968
89,152
92,336
95,520
98,704
101,888
Shared Memory Per Block

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	102,393.92	Average L1 Active Cycles [cycle]	102,393.92
Average L2 Active Cycles [cycle]	67,886.12	Average SMSP Active Cycles [cycle]	101,650.58
Average DRAM Active Cycles [cycle]	49,316	Total SM Elapsed Cycles [cycle]	2,550,280
Total L1 Elapsed Cycles [cycle]	2,550,280	Total L2 Elapsed Cycles [cycle]	1,693,360
Total SMSP Elapsed Cycles [cycle]	10,201,120	Total DRAM Elapsed Cycles [cycle]	2,100,224

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	102,393.92	100,556	104,336	2,457,454
SMSP Active Cycles	101,650.58	99,512	104,250	9,758,456
L1 Active Cycles	102,393.92	100,556	104,336	2,457,454
L2 Active Cycles	67,886.12	66,290	69,811	1,086,178
DRAM Active Cycles	49,316	49,280	49,392	197,264

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	363,520	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.09	Avg. Divergent Branches [branches]	0

Uncoalesced Global Accesses
Est. Speedup: 6.41%
This kernel has uncoalesced global accesses resulting in a total of 32768 excessive sectors (10% of the total 327680 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide has additional information on reducing uncoalesced device memory accesses.

Key Performance Indicators

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
0x79d441272de0 in gemm_wmma_kernel	4,096	13
0x79d441272dd0 in gemm_wmma_kernel	4,096	13
0x79d441272dc0 in gemm_wmma_kernel	4,096	13
0x79d441272db0 in gemm_wmma_kernel	4,096	13
0x79d441272bb0 in gemm_wmma_kernel	4,096	13

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
0x79d4412728a0 in gemm_wmma_ke...	571	24
0x79d441272720 in gemm_wmma_ke...	449	19
0x79d441272a40 in gemm_wmma_ke...	243	10
0x79d4412729d0 in gemm_wmma_ke...	152	6
0x79d4412728e0 in gemm_wmma_ke...	118	5

Most Instructions Executed

Location	Value	Value (%)
0x79d4412728b0 in gemm_wmma_ke...	65,536	2
0x79d4412728a0 in gemm_wmma_ke...	65,536	2
0x79d441272890 in gemm_wmma_ke...	65,536	2
0x79d441272880 in gemm_wmma_ke...	65,536	2
0x79d441272870 in gemm_wmma_ke...	65,536	2

Follow the *rules outputs* to get guidance on how to navigate through the report and quickly discover performance bottlenecks in this kernel.
You could also disable individual sections to focus on selected performance aspects and make profiling faster.